

# Create, expand, factorise, rearrange and simplify linear expressions, applying the associative, commutative, identity, distributive and inverse properties

## Learning intention

- create, expand, factorise, rearrange and simplify linear expressions, applying the associative, commutative, identity, distributive and inverse properties.

## Teach and model

- using the distributive, associative, commutative, identity and inverse properties to expand and factorise algebraic expressions using strategies such as the area model.
- rearranging and simplifying linear expressions involving variables with integer coefficients and constants; using manipulatives such as algebra tiles to support calculations.
- demonstrating the relationship between factorising and expanding linear expressions using manipulatives, such as algebra tiles or area models, and describing with mathematical language.

## Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

## Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

## Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.